RG Topology: The Resolution Interior

A Path from Experiments to Topology Beside the Star Construction

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Abstract

The star construction reaches topology from algebra. A commutative C*-algebra determines a compact Hausdorff space, its Gelfand spectrum. We give a path that starts elsewhere, from *experiments*: repeatable protocols with finite records and declared error control. It reaches topology through one operator, the *resolution interior* $\mathfrak{r}_{\mathscr{P}}(A)$, the set of states at which the proposition A can be verified by the protocol family $\mathscr{P}$.

What we prove.

(i) From four operational axioms, the verifiable propositions are closed under finite conjunction and countable disjunction. Together with a characterization theorem, $\mathfrak{r}_{\mathscr{P}}$ is a Kuratowski interior operator, and hence a topology $\tau_{\mathscr{P}}$.

(ii) For repeatable finite-outcome protocols, $\tau_{\mathscr{P}}$ is the pullback of the weak topology on record laws. It *contains* a law-neighbourhood exactly where verification succeeds.

(iii) The Kolmogorov (T_0) quotient of $\tau_{\mathscr{P}}$ is RG's observational quotient: topological indistinguishability *is* ignorance. For countable noisy families this quotient is metrizable.

(iv) $E \mapsto \tau_E$ is a monotone functor from the Blackwell order to topologies. It is invariant under lossless re-descriptions, but not faithful.

(v) One-sided (noise-free) protocols give non-Hausdorff T_0 topologies. The arrival detector is Sierpiński space, and exact onset is lower semicontinuous.

(vi) Reach: every second-countable T_0 space is the resolution topology of a countable family of one-sided protocols. Every separable metrizable space is that of a countable family of noisy coins. Conversely, every topology produced by a countable protocol family is second countable.

(vii) Where both paths apply (commutative, Hausdorff), the two meet: Gelfand duality applied to the expectation functionals of the protocols reconstructs the same quotient.

Scope. The frameworks behind this path (Kuratowski, topology via logic, statistical verifiability, Urysohn embedding) are recognized. The contribution is the derivation from RG's protocol axioms, the identification of the Kolmogorov quotient with ignorance, the functor from the Blackwell order, and the physical reach statement. Seven exact witnesses check constants. No claim rests on sampling.

Contents

1 Two paths to topology 2

2 Operational setting 2

3 The resolution interior is a topology **3**

4 Structure theorems **4**

5 Reach **5**

6 Where the two paths meet, and where they differ **5**

7 The RG symbol: domain, purpose, limits **6**

8 Witnesses and limits **6**

1 Two paths to topology

Imported Theorem 1.1 (The star path; Gelfand–Naimark [1])**.** *Every commutative unital C^*-algebra $\mathcal{A}$ is isometrically $*$-isomorphic to $C(\operatorname{Spec}\mathcal{A})$, and $\operatorname{Spec}\mathcal{A}$ is compact Hausdorff. This is a contravariant equivalence between commutative unital C^*-algebras and compact Hausdorff spaces.*

The star path starts from an *algebra* with an involution and reaches its spectrum. Its reach is the compact (or locally compact) Hausdorff spaces. RG's question is different. Given only experiments, with finite records, repeatability and error control, *why must a topology appear at all, which one, and how far can such a path reach?*

2 Operational setting

Assumption 2.1 (Protocol axioms)**.** A protocol family $\mathscr{P}$ on a state set X consists of protocols p. Each assigns to $x \in X$ a law $P_x^{(p)}$ on a finite outcome set, and can be repeated independently.

(O1) *Finite records.* Every decision uses finitely many outcomes.

(O2) *Parallel composition.* Independent runs of protocols in $\mathscr{P}$ can be combined.

(O3) *Declared error.* A verifier is admissible at level α if its false-acceptance probability is at most α at every state where the proposition is false.

(O4) *Abstention.* A verifier may decline to accept. It is never forced to decide.

Definition 2.2 (Verifiability; resolution interior)**.** A proposition $A \subseteq X$ is *verifiable* by $\mathscr{P}$ if, for every $\alpha > 0$, there are verifiers φ_n using n records such that:

- $\sup_{y \notin A} P_y(\varphi_n \text{ accepts}) \le \alpha$ for all n;
- $P_x(\varphi_n \text{ accepts}) \to 1$ for every $x \in A$.

Write $\mathcal{V}_{\mathscr{P}}$ for the verifiable propositions. The *resolution interior* is $\mathfrak{r}_{\mathscr{P}}(A) = \bigcup\{V \in \mathcal{V}_{\mathscr{P}} : V \subseteq A\}$.

The definition follows the notion of statistical verifiability of Genin and Kelly [2]. The verifier must never be fooled outside A more than the declared rate, and must eventually confirm A wherever it holds. This is exactly the asymmetry of the wall-tap onset: arrival can be confirmed; non-arrival is never confirmed.

3 The resolution interior is a topology

Lemma 3.1 (Finite conjunction)**.** *If $A, B \in \mathcal{V}_{\mathscr{P}}$, then $A \cap B \in \mathcal{V}_{\mathscr{P}}$.*

Proof. Split the records (O2), run the two verifiers at levels α, and accept iff both accept. If $y \notin A \cap B$, then $y \notin A$ or $y \notin B$, so the false acceptance is at most α. On $A \cap B$, the acceptance probability is at least $P(\varphi^A \text{ acc}) + P(\varphi^B \text{ acc}) - 1 \to 1$. □

Lemma 3.2 (Countable disjunction)**.** *If $A_1, A_2, \ldots \in \mathcal{V}_{\mathscr{P}}$, then $\bigcup_i A_i \in \mathcal{V}_{\mathscr{P}}$.*

Proof. At budget n, run the verifiers for $A_1, \ldots, A_{k(n)}$ on disjoint blocks of $\lfloor n/k(n) \rfloor$ records, with $k(n) \to \infty$ slowly. Use level $\alpha 2^{-i}$ for A_i, and accept if any accepts. Outside the union, the false acceptance is at most $\sum_i \alpha 2^{-i} \le \alpha$. At $x \in A_j$, once $k(n) \ge j$ the j-th block grows without bound, so it accepts with probability tending to 1. □

Theorem 3.3 (Characterization for finite-outcome protocols)**.** *Let $\mathscr{P} = \{p_1, p_2, \ldots\}$ be countable, with finite outcome sets. Let $\Lambda : X \to \prod_k \Delta_{m_k}$ be the law map, where the product of simplices carries the product of Euclidean topologies (equivalently, of total variation). Then*

$$\mathcal{V}_{\mathscr{P}} = \{\Lambda^{-1}(O) : O \text{ open}\}.$$

Proof. Necessity. For a verifier with n records, a coupling argument gives

$$|P_x(\varphi_n \text{ acc}) - P_y(\varphi_n \text{ acc})| \le n \operatorname{TV}(P_x^{\otimes}, P_y^{\otimes}) \le n \textstyle\sum_k \operatorname{TV}(P_x^{(p_k)}, P_y^{(p_k)})$$

over the protocols used. Fix $x \in A$ and n with $P_x(\varphi_n \text{ acc}) > \alpha + \delta$. Every y whose laws, on the finitely many protocols used, are within δ/n of those of x has acceptance above α, so it lies in A. Hence A contains a basic open neighbourhood of $\Lambda(x)$.

Sufficiency. Basic opens are finite intersections of preimages of open sets of single simplices (Lemma 3.1). For one protocol with m outcomes and open $O \subseteq \Delta_m$, accept iff the ℓ^∞-ball of radius $\rho_n = \sqrt{\log(2m/\alpha)/(2n)}$ around the empirical law lies inside O. By Hoeffding and a union bound, the true law lies in that ball with probability at least $1 - \alpha$, uniformly. So acceptance at $y \notin \Lambda^{-1}(O)$ requires a deviation event of probability at most α. At $x \in \Lambda^{-1}(O)$, $\rho_n \to 0$ and the empirical law converges, so acceptance tends to 1. Arbitrary opens are countable unions of basic opens (second countability), and Lemma 3.2 applies. □

Witnesses. T1: for $U = (0.3, 0.7)$, worst false acceptance $\le 0.0015 \le \alpha$ at $n = 50, \ldots, 3200$, and acceptance at 0.5 rising $0.11 \to 0.996 \to 1$. T2: the Lipschitz bound, checked exactly.

Theorem 3.4 (Resolution interior)**.** *Under Assumption 2.1 and the hypotheses of Theorem 3.3, $\mathcal{V}_{\mathscr{P}}$ is a topology $\tau_{\mathscr{P}}$, and $\mathfrak{r}_{\mathscr{P}}$ is its interior operator. That is, $\mathfrak{r}(X) = X$, $\mathfrak{r}(A) \subseteq A$, $\mathfrak{r}(\mathfrak{r}(A)) = \mathfrak{r}(A)$ and $\mathfrak{r}(A \cap B) = \mathfrak{r}(A) \cap \mathfrak{r}(B)$, which are Kuratowski's axioms.*

Proof. By Theorem 3.3, $\mathcal{V}_{\mathscr{P}}$ is the initial topology of Λ. $\mathfrak{r}$ is the union of open subsets, i.e. the interior. The axioms follow, and Lemmas 3.1–3.2 give the operational content of the intersection and union clauses. □

Remark 3.5 (Why a topology appears at all). Two operational facts force it. Finite records make acceptance probabilities continuous in the law (necessity), and repeatable protocols make empirical laws converge (sufficiency). Topology is what remains of an experiment's distinguishing power once rates and constants are forgotten.

4 Structure theorems

Theorem 4.1 (The Kolmogorov quotient is the ignorance quotient)**.** *x and y are topologically indistinguishable in $\tau_{\mathscr{P}}$ iff $P_x^{(p)} = P_y^{(p)}$ for every $p \in \mathscr{P}$. Hence the T_0 quotient of $(X, \tau_{\mathscr{P}})$ is RG's observational quotient $X/\!\sim_{\mathscr{P}}$. For a countable family it is homeomorphic to the law image, which is metrizable.*

Proof. Initial topologies identify exactly the points with equal images, and the law image sits in a metrizable product. Witness: T3, where the four states fall into the three law classes, which are exactly the T_0 classes. □

Theorem 4.2 (A monotone functor from the Blackwell order)**.** *(a) If F is a garbling of E (on the same state set), then $\tau_F \subseteq \tau_E$.*

(b) Lossless re-descriptions leave τ unchanged (descent).

(c) Joining protocol families gives the join of topologies.

(d) The functor is not faithful: equal topologies do not imply Blackwell equivalence.

Proof. (a) A verifier for F, applied to records garbled by the kernel, is a verifier for E. (b) Apply (a) in both directions. (c) This holds for initial topologies of product maps. (d) Witness T5: Bernoulli($\sin^2 \frac{\theta}{2}$) and two i.i.d. Bernoulli($\sin^2 \frac{\theta}{2\sqrt{2}}$) both embed $(0, \pi)$ homeomorphically, so they give the same topology. Their Hellinger affinities at separation 1 are 0.8776 and 0.8801, so they are not equivalent. □

Reading. The topology is the coarsest invariant of experiments: it forgets rates, constants and efficiency. The finer invariants are those of RG's other layers: the Le Cam metric, Fisher geometry and RLCT rates.

Theorem 4.3 (One-sided protocols and the specialization order)**.** *Suppose a protocol is noise-free and one-sided: it outputs "detected" only when a property holds, and outputs it with probability tending to 1 when the property holds. Then the detected property is verifiable, but its complement need not be.*

- *The arrival detector gives Sierpiński space $\{not \leq arrived\}$, which is T_0 but not T_1 (T4).*
- *The specialization preorder ($x \leq y$ iff every verifiable property of x holds at y) is RG's "verifiably at least as much".*
- *The exact onset is lower semicontinuous, and not continuous. This is the structural reason for the onset fragility in the wall-tap print.*

Corollary 4.4 (Hidden and null, topologically)**.**

- *A distinction is* hidden *for $\mathscr{P}$ if the points are indistinguishable in $\tau_{\mathscr{P}}$ but separated in $\tau_{\mathscr{P}'}$ for an admissible refinement $\mathscr{P}'$.*
- *It is* null *if no admissible refinement separates them.*
- *The fold of the wall-tap overlap protocol is hidden: b and $2d - b$ are indistinguishable in τ_{P_3} and separated in τ_{P_4} (T6). The T_0 quotient of τ_{P_3} is the orbifold $|b - d|$.*

5 Reach

Theorem 5.1 (Reach of the resolution path)**.** *(a) Every topology $\tau_{\mathscr{P}}$ from a countable protocol family is second countable.*

(b) Every second-countable T_0 space is homeomorphic to $(X, \tau_{\mathscr{P}})$ for a countable family of one-sided protocols.

(c) Every separable metrizable space is homeomorphic to $(X, \tau_{\mathscr{P}})$ for a countable family of noisy two-outcome protocols (coins).

Proof. (a) Rational balls in a countable product of simplices form a countable base, and preimages preserve this.

(b) Let $\{B_k\}$ be a countable base. The map $x \mapsto (\mathbf{1}_{B_k}(x))_k$ is an embedding into the Sierpiński cube $\mathbb{S}^{\mathbb{N}}$, a classical result [5]. Realize each coordinate by a one-sided detector of B_k.

(c) By Urysohn's metrization theorem, such a space embeds in the Hilbert cube $[0,1]^{\mathbb{N}}$. Realize coordinate k by a coin with bias $f_k(x)$. The coin's law topology is the Euclidean topology on the bias.

Witnesses: T7 recovers a three-point non-Hausdorff chain exactly from two one-sided detectors, and embeds the circle via two coins $\big((1+\cos\varphi)/2, (1+\sin\varphi)/2\big)$. □

Remark 5.2 (A physical reading of second countability). Part (a) says that any topology reachable by a countable family of finite-record experiments is second countable. Non-second-countable spaces, such as uncountable products, are not reachable by physical observation of this kind. They enter only as declarations.

6 Where the two paths meet, and where they differ

Theorem 6.1 (Meeting point with the star path)**.** *Let $\mathscr{P}$ be countable with finite outcomes, and let $\overline{\Lambda(X)}$ be the closure of the law image, which is compact. The unital C^*-algebra generated by the expectation functionals $x \mapsto \mathbb{E}_x^{(p)}[f]$ is the algebra of continuous functions on $\overline{\Lambda(X)}$. Its Gelfand spectrum is $\overline{\Lambda(X)}$, a compactification of the T_0 quotient of $\tau_{\mathscr{P}}$.*

Proof. The coordinate functions of Λ are expectation functionals, and they separate points of $\overline{\Lambda(X)}$. Stone–Weierstrass gives density, and Gelfand–Naimark identifies the spectrum. □

	Star path (Gelfand–Naimark)	Resolution path ($\mathfrak{r}_{\mathscr{P}}$)
Starts from	an algebra with involution	experiments: finite records, repetition, declared error
Primitive	characters of the algebra	verifiable propositions
Reaches	compact (or locally compact) Hausdorff spaces	every second-countable T_0 space; metrizable for noisy families
Non-Hausdorff spaces	no	yes (one-sided protocols; Sierpiński)
Quotient	points are characters	T_0 quotient = observational quotient (ignorance)
Order	none intrinsic	monotone functor from the Blackwell order
Finer layers	norm, states	Le Cam metric, Fisher geometry, RLCT rates

	Star path (Gelfand–Naimark)	Resolution path ($\mathfrak{r}_{\mathscr{P}}$)
Noncommutative	noncommutative geometry	quantum protocols give the weak* topology on states (the two paths meet again)

7 The RG symbol: domain, purpose, limits

Symbol. $\mathfrak{r}_{\mathscr{P}}(A)$, the resolution interior: "the part of A that the protocol family can confirm".

Domain. Any state set with a declared countable family of repeatable finite-outcome protocols (O1–O4). Dominated continuous outcomes reduce to this case by finite partitions (Čencov print, Section 12).

What it reaches.

- Topology on states, from experiments alone.
- The T_0 quotient as ignorance.
- Non-Hausdorff topologies from one-sided detectors.
- Every second-countable T_0 space.

Purpose.

- Give experimental science a topology it can *measure into existence*: open means confirmable.
- Make "hidden" and "null" topological notions.
- Order topologies by the Blackwell order of the experiments that produce them.

Limits.

- *Not new mathematics of topology.* Kuratowski interiors, topology via logic (Vickers, Smyth, Abramsky [3, 4]), statistical verifiability (Genin–Kelly [2]) and Urysohn embedding are established. RG's contribution is the derivation from its protocol axioms and the identifications above.
- *Countable families give only second-countable spaces.*
- *Repetition is declared:* non-i.i.d. records need their own concentration results.
- *The topology forgets rates.* It does not replace the Le Cam, Fisher or RLCT layers.

8 Witnesses and limits

- `rg_topology_witnesses.py` runs T1–T7. They are exact binomial sums, symbolic or exact finite computations, with no sampling. Reruns are byte-identical.
- *Limits.* The statements are proved for countable families of finite-outcome protocols. Imported results are Gelfand–Naimark, Stone–Weierstrass, Urysohn, the Sierpiński-cube embedding, and Hoeffding's inequality. The T1 soundness check uses a grid of false parameters; the proof covers all of them.

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